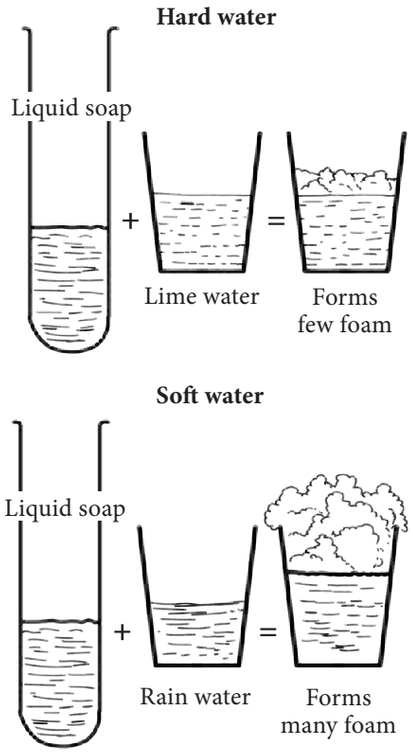


Figure 13.7 Hard water and soft water forming foam

it is called soft water. In this water, soaps or detergents form lather easily (Fig. 13.7).

Sometimes, minerals and salts are present in water in such a large quantity that soaps or detergents form a thick precipitate called scum instead of forming lather. This makes the removal of dirt further difficult. Such water is called hard water. Hardness of water is due to the presence of dissolved salts of calcium and magnesium. Hardness may be temporary or permanent. Temporary hardness is due to the presence of carbonate and bicarbonate salts of calcium and magnesium, and permanent hardness results due to the presence of chloride and sulphate salts of calcium and magnesium.

a. Disadvantages of hard water

- It is not good for washing clothes. It forms scum with soap and detergents, which makes the soap ineffective and also spoils the clothes further.
- It damages the utensils and containers in which it is stored and forms a hard layer.
- It forms scales on the machine parts used in industries and decreases their efficiency (Fig. 13.8).
- It results in stomach ailments if consumed for a long period.



Figure 13.8 Scales on the machine pipes

Activity 6

Take samples of water from different sources (like a tube well, a lake, a pond or a river) and pour equal quantities of each sample of water into different test tubes. Measure the height of water in each test tube with a scale. Add one or two drops of liquid soap to each test tube.

Shake each test tube five times and observe the height of the lather in each sample. Record your observations in the table. Which water is soft? Which water is hard? Can you say why?

Samples of water (Source)	Height of lather
Tap water	
Well water	
Pond water	
River water	

b. Removal of hardness

Different methods are followed to remove the hardness from water depending on whether it is temporary hardness or permanent hardness. Some of them are explained below.

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